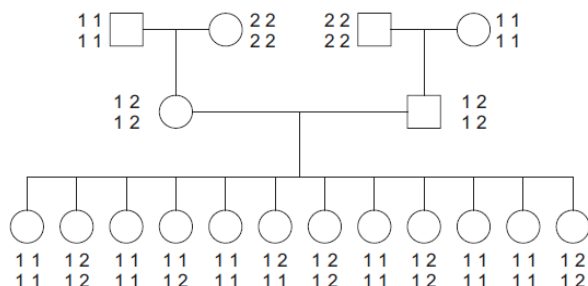


P8139
Homework #2
Due 03/03/26

1. Consider the pedigree displayed in the figure below.
 - (a) How many informative meioses does the pedigree have?
 - (b) Indicate recombinants and non-recombinants
 - (c) Estimate the recombination fraction in this pedigree.
 - (d) Determine the LOD score

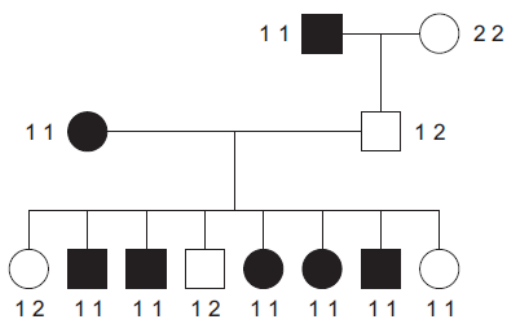


Answer:

(a)24; (c)1/12

2. Consider a rare autosomal recessive disease segregating with complete penetrance and no phenocopies in the family shown in the figure below. All family members have been genotyped at all genetic markers.

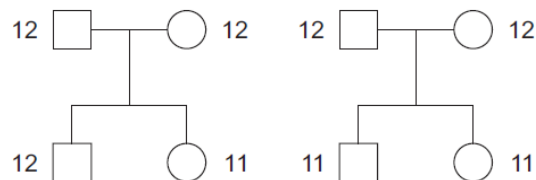
- (a) Determine the parental haplotypes.
- (b) How many recombination events have occurred?
- (c) Determine the LOD score



Answer:

(b)1/8

3. Determine the IBD and IBS values of the sib-pairs in the two pedigrees shown below.



4. Read the paper “Three lectures on case-control genetic association analysis” published in *Briefings in Bioinformatics*, 9(1): 1-13, 2007 and write a half-page paragraph with your understanding. Going through the R code in the paper and reproduce all numbers.